

Shortly Scrapers

Working document for the daily news scraper, summarizer, corporate case agent, and topic agents.

Document purpose	Explain how content is fetched, cleaned, summarized, stored, reviewed, and finally used in email digests.
Project	Shortly AI Emailer
Primary runtime	Supabase Edge Functions + pg_cron + OpenAI + Supabase Postgres
Prepared date	19 June 2026

This PDF is based on the current code in the repository. It avoids secrets and describes behavior, schedules, failure modes, and maintenance points.

1. Executive Summary

The Shortly scraping system has four content-producing paths. The daily news path fetches RSS feeds from TOI, Economic Times, and The Hindu, stores unique articles as pending, then summarizes and ranks them for QA. Separate editorial agents create deeper topic drafts and corporate case studies. A human editor approves content in the dashboard before it is used in emails.

1. Fetch RSS, HTML listings, or Reddit JSON	2. Clean Strip HTML, social prompts, phone lines, page furniture	3. Select Dedupe, freshness filter, source weighting	4. Generate OpenAI summaries, drafts, case studies	5. Review QA approves, edits, rejects, or sends
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The main operational design choice is safety through staged status changes. Raw scraped items are not emailed directly. They move through pending, summarized, approved, and sent states. Topic agents write into separate draft tables, keeping long-form editorial content separate from the daily news queue.

2. Scheduled Pipeline

Scheduling is handled by Supabase pg_cron. Times in the SQL file are UTC; India Standard Time is UTC+05:30.

IST time	UTC cron	Job name	Function	Purpose
08:00	30 2 * * *	shortly-scrape-8am-ist	scrape-news	Fetch daily news RSS items and insert pending rows.
08:15	45 2 * * *	shortly-summarize-815am-ist	summarize-articles	Summarize recent pending items and prepare QA queue.
09:00	30 3 * * *	shortly-send-digest-9am-ist	send-daily-digest	Send the approved daily digest. Protected by a once-per-IST-day scheduled-send guard.
10:00	30 4 * * 1-5	shortly-corporate-case-weekdays	corporate-case-agent	Draft one source-backed corporate case study on weekdays.
10:10	40 4 * * 1-6	shortly-real-estate-mon-sat	editorial-topic-agent	Draft Real Estate topic issue.
10:20	50 4 * * 1-6	shortly-policy-partner-mon-sat	editorial-topic-agent	Draft Policy Partner issue.
10:30	0 5 * * 1-6	shortly-money-matters-mon-sat	editorial-topic-agent	Draft Money Matters issue.
10:40	10 5 * * 1-6	shortly-wellness-daily-mon-sat	editorial-topic-agent	Draft Wellness Daily issue.

The cron helper function reads the service role key from Supabase Vault and invokes Edge Functions through pg_net with longer timeouts for summarization and sending.

3. Daily News Scraper

Function: `supabase/functions/scrape-news/index.ts`

This function is the fast RSS ingest layer. It intentionally uses a small source list so fetch time stays predictable and the queue is easier for QA to control.

Source	RSS URL	Weight	Topic
TOI	timesofindia.indiatimes.com/rssfeedstopstories.cms	1.00	India
ET	economictimes.indiatimes.com/rssfeedstopstories.cms	0.95	India Business
The Hindu	thehindu.com/news/national/feeder/default.rss	1.00	India

Daily scraper step-by-step:

- Fetch every configured RSS source in parallel with a Shortly user agent.
- Parse RSS or Atom items using the shared lightweight parser.
- Clean the description field to remove HTML, app prompts, social calls, phone-only lines, subscription text, and page furniture.
- Create article rows with title, URL, raw content, source, topic, source weight, and status pending.
- Dedupe within the scrape batch by URL.
- Upsert into `public.articles` on URL with `ignoreDuplicates`, so re-running the scraper does not create duplicates.

Stage	Input	Output	Status written
scrape-news	RSS feeds	Article candidates	pending
dedupe	Candidate URLs	Unique URL rows	pending
upsert	Unique candidates	<code>public.articles</code>	pending

4. Text Cleaning and RSS Parsing

Shared files: `_shared/article-text.ts` and `_shared/rss.ts`

Cleaning is important because RSS descriptions often include unrelated page furniture. The cleaner removes inline noise like follow prompts, Telegram or WhatsApp joins, app downloads, ads, links, read more text, and subscription prompts. It also removes whole lines that look like contact information, helplines, copyright text, live-blog furniture, galleries, or phone-only lines.

Problem	How the code handles it
HTML in RSS excerpts	Scripts, styles, tags, paragraph breaks, and entities are stripped or decoded.
Social and newsletter prompts	Regex patterns remove follow, join, subscribe, download app, click here, read more, and share lines.
Phone numbers and contact lines	Phone-only lines and lines containing helpline, WhatsApp, Telegram, contact us, customer care, etc. are filtered.
Short or noisy excerpts	<code>summarize-articles</code> can fetch the full article page and clean the body/article HTML.
RSS versus Atom	The parser supports RSS item and Atom entry blocks, using title, link/guid, description/summary/content, and date tags.

5. Summarization and QA Queue

Function: `supabase/functions/summarize-articles/index.ts`

The summarizer turns pending rows into editor-ready cards. It only looks at recent pending articles from the last 10 hours, ordered by source weight and freshness, and processes up to 50.

OpenAI prompt contract:

- Write exactly three sentences.
- Target 60-90 words total.
- Use active voice and preserve concrete names, dates, numbers, percentages, and currencies.
- Classify each article as wrapped or ahead.
- Rate prominence from 1 to 5.
- Return strict JSON: summary, section, prominence.

Ranking formula:

$$\text{score} = \text{source_weight} * 0.4 + \text{prominence_normalized} * 0.3 + \text{freshness} * 0.3$$

After successful summarization, articles are updated with summary, section, prominence, rank_score, summarized_at, and status summarized. The dashboard shows these summarized rows to QA. The best 50 remain summarized; overflow rows can be demoted back to pending.

Stage	Input	Output	Status written
summarize-articles	pending articles from last 10 hours	3-sentence summary plus metadata	summarized
OpenAI JSON parse	title, source, URL, cleaned excerpt	summary, section, prominence	summarized or failed
ranking	source weight, prominence, freshness	rank_score	summarized

6. QA Review and Daily Email Selection

The scraper does not decide final email content by itself. QA can approve, reject, edit headlines, edit summaries, and reorder approved content. Daily Wrap approval is capped at 10 for the normal daily email flow.

If QA does not explicitly select enough articles, the send function can fill remaining slots from approved/summarized articles by rank. This fallback is useful operationally, but it is separate from scraping: the scraper only creates the queue.

The scheduled 9 AM send path now uses a once-per-IST-day guard. If the scheduler retries or invokes the function multiple times, only the first scheduled digest of that IST day should send.

7. Corporate Case Agent

Function: `supabase/functions/corporate-case-agent/index.ts`

This is not a simple news scraper. It is a source-backed case-study builder. It scans Indian business sources, selects one company-focused story, extracts public article text, asks OpenAI to draft a 400-600 word Corporate Case, validates the structure, and stores it for editor review.

Source	Use
The Ken	High-signal business analysis feed.
Inc42	Startup and tech-business candidates.
Moneycontrol business feed	Indian business reporting, filtered to business URLs.
ET Prime	Premium business feed candidates.

Selection filters and safeguards:

- Rejects personal finance, tax advice, deposits, insurance, home loans, stock tips, and generic macro stories.
- Requires a named company, enough public source text, and a business-model question.
- Avoids repeating cases from the previous 14 days when possible.
- Writes summary, detail, bull case, bear case, open question, comparison or analogy, inference notes, and editor checklist.
- Validates word counts and rejects weak drafts that do not meet structure.

Stage	Input	Output	Status written
collectCandidates	Corporate RSS feeds	company case candidates	not stored yet
rankCandidates	candidate list	ranked URLs with company and case type	not stored yet
writeCase	selected article text	case-study draft	draft
editor action	draft case	approved/rejected/updated case	approved, rejected, draft

8. Editorial Topic Agent

Function: `supabase/functions/editorial-topic-agent/index.ts`

The editorial topic agent builds specialized newsletters that are stored in `public.editorial_drafts`. It supports both single-article style topics and hybrid formats with five briefs plus one feature.

Topic	Format	Core source types	Output
Industry: Real Estate	Hybrid	ET Realty, Moneycontrol Real Estate	Five briefs plus one thesis.
Policy Partner	Single	PRS, Bar & Bench, LiveLaw, RBI, explainers	One policy explainer issue.
Money Matters	Hybrid	Mint, Moneycontrol, RBI, SEBI, finance sources, Reddit compass	Five briefs plus one take.
The Wellness Daily	Single	Washington Post Wellness, Guardian, HBR, Reddit compass	One wellness advisory.

How it works:

- Collects candidates from RSS, HTML listings, and Reddit JSON depending on topic source config.
- Filters stale candidates older than five days when publication dates are available.
- Uses Reddit as editorial compass only; Reddit is not cited as evidence.
- Uses OpenAI to select evidence URLs from up to 70 candidates.
- Fetches readable article text for selected cited evidence.
- Writes a source-bound draft in the topic's required structure and voice.
- Runs validation, repair, expansion, and safety-line enforcement before storing the draft.

9. Database Tables and Status Flow

Table	Created by	Purpose
articles	scrape-news, summarize-articles, review-article	Daily news queue with raw content, summary, section, rank, status, and timestamps.
corporate_cases	corporate-case-agent	Source-backed business case-study drafts and review state.
editorial_drafts	editorial-topic-agent	Topic newsletter drafts with content JSON, source links, checklist, and review state.
subscribers	dashboard/subscriber API	Recipients and topic preferences.
digests	send functions	One row per digest send attempt with article IDs and delivery counts.
article_deliveries	send functions	Per-recipient delivery log with provider status and errors.

pending RSS item inserted	summarized AI summary ready	approved QA selected	sent Email sent
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Corporate cases and editorial drafts use draft, approved, rejected, and published-style statuses separately from the daily article pipeline.

10. Reliability, Idempotency, and Failure Handling

Area	Protection
Duplicate RSS items	URL-based in-batch dedupe plus database upsert on URL with ignoreDuplicates.
Noisy RSS excerpts	Shared cleaner removes contact lines, prompts, links, ads, copyright, and phone-only lines.
Short/incomplete excerpts	Summarizer can fetch full public article text when the RSS excerpt is too short or noisy.
Slow source fetches	Sources are fetched in parallel; topic jobs are staggered by cron.
OpenAI failures	Per-article failures are captured; successful summaries still persist.
Duplicate scheduled sends	Scheduled digest path checks for an existing digest in the current IST day before sending.
Source errors	Functions return sourceErrors arrays to aid debugging.

Important operational note: `pg_net` may report '1 row' when a request is queued successfully. To verify deeper function success, check the function logs, digest rows, `article_deliveries`, and returned payloads where available.

11. Operator Runbook

Daily check sequence:

- After 08:00 IST: confirm scrape-news inserted new pending rows or returned zero with source errors explained.
- After 08:15 IST: confirm summarize-articles moved rows to summarized and that failures are not widespread.
- Before 09:00 IST: QA should approve or edit the desired Daily Wrap articles.
- After 09:00 IST: check digests and article_deliveries for sent/failed counts.
- After topic-agent runs: review corporate_cases and editorial_drafts for draft quality and source links.

Useful SQL checks:

```
select status, count(*) from public.articles group by status;
select sent_at at time zone 'Asia/Kolkata' as sent_at_ist, recipients, sent, failed from
public.digests order by sent_at desc limit 10;
select jobid, jobname, schedule, active, command from cron.job order by jobid;
```

When scraping looks wrong:

- Check whether source RSS URLs still return HTTP 200.
- Look at sourceErrors from scrape-news or the topic agent.
- Inspect raw_content for page furniture or phone lines if summaries look noisy.
- If no articles appear, confirm the source list and accepts filters are not too strict.
- If summaries are too short, inspect OpenAI failures and whether readable page fetch is blocked by the publisher.

12. Maintenance Points

Change needed	File to edit	What to watch
Add/remove daily news source	_shared/sources.ts	Keep source count small; set sensible weight and topic.
Tune summary style	summarize-articles/index.ts	Update system prompt and max_tokens carefully.
Reduce noisy content	_shared/article-text.ts	Add noise regex patterns and test against real excerpts.
Change topic sources	_shared/editorial-topics.ts	Update source config, accepts filters, and topic rules.
Change corporate sources	_shared/corporate-case-sources.ts	Filter out finance advice and weak company stories.
Change schedules	supabase/cron.sql	Remember cron times are UTC, not IST.

The safest maintenance approach is to update source configs first, run the function manually, inspect returned counts and sourceErrors, then deploy/schedule only after the QA queue looks clean.

End of document